

Beginning MATLAB and Simulink

From Novice to Professional

Sulaymon Eshkabilov

Apress®

Beginning MATLAB and Simulink: From Novice to Professional

Sulaymon Eshkabilov
Fargo, United States

ISBN-13 (pbk): 978-1-4842-5060-0

ISBN-13 (electronic): 978-1-4842-5061-7

<https://doi.org/10.1007/978-1-4842-5061-7>

Copyright © 2019 by Sulaymon Eshkabilov

This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

Trademarked names, logos, and images may appear in this book. Rather than use a trademark symbol with every occurrence of a trademarked name, logo, or image we use the names, logos, and images only in an editorial fashion and to the benefit of the trademark owner, with no intention of infringement of the trademark.

The use in this publication of trade names, trademarks, service marks, and similar terms, even if they are not identified as such, is not to be taken as an expression of opinion as to whether or not they are subject to proprietary rights.

While the advice and information in this book are believed to be true and accurate at the date of publication, neither the authors nor the editors nor the publisher can accept any legal responsibility for any errors or omissions that may be made. The publisher makes no warranty, express or implied, with respect to the material contained herein.

Managing Director, Apress Media LLC: Welmoed Spahr

Acquisitions Editor: Steve Anglin

Development Editor: Matthew Moodie

Coordinating Editor: Mark Powers

Cover designed by eStudioCalamar

Cover image from rawpixel.com

Distributed to the book trade worldwide by Springer Science+Business Media New York, 233 Spring Street, 6th Floor, New York, NY 10013. Phone 1-800-SPRINGER, fax (201) 348-4505, e-mail orders-ny@springer-sbm.com, or visit www.springeronline.com. Apress Media, LLC is a California LLC and the sole member (owner) is Springer Science + Business Media Finance Inc (SSBM Finance Inc). SSBM Finance Inc is a **Delaware** corporation.

For information on translations, please e-mail editorial@apress.com; for reprint, paperback, or audio rights, please email bookpermissions@springernature.com.

Apress titles may be purchased in bulk for academic, corporate, or promotional use. eBook versions and licenses are also available for most titles. For more information, reference our Print and eBook Bulk Sales web page at <http://www.apress.com/bulk-sales>.

Any source code or other supplementary material referenced by the author in this book is available to readers on GitHub via the book's product page, located at www.apress.com/9781484250600. For more detailed information, please visit <http://www.apress.com/source-code>.

Printed on acid-free paper

To the memory of my father.

To my mother.

To my wife, Nigora, after 25 wonderful years together.

Table of Contents

About the Author	xvii
About the Technical Reviewers	xix
Acknowledgments	xxi
Introduction	xxiii
Chapter 1: Introduction to MATLAB	1
MATLAB's Menu Panel and Help	1
The MATLAB Environment.....	7
Working in the Command Window	9
Command Window and Variables.....	11
Using Variables	11
When to Use the Command Window	18
Different Variables and Data Sets in MATLAB	23
Numerical Data/Arrays	25
NaN (Not-a-Number).....	33
Character Types of Variables	38
Function Handle.....	40
Logical Arrays	44
Table Arrays	48
Cell Arrays	50
Structure Arrays	54
Complex Numbers	59
Precision.....	59
M-File and MLX-File Editors	60
M-File Editor	60
MLX-File Editor	62

TABLE OF CONTENTS

Closing the MATLAB Window 72

Summary..... 72

References..... 72

Exercises for Self-Testing 73

 Exercise 1 73

 Exercise 2 73

 Exercise 3 73

 Exercise 4 74

 Exercise 5 74

 Exercise 6 74

 Exercise 7 74

 Exercise 8 74

 Exercise 9 74

 Exercise 10 75

 Exercise 11 75

 Exercise 12 75

 Exercise 13 76

 Exercise 14 76

 Exercise 15 76

 Exercise 16 77

 Exercise 17 77

 Exercise 18 77

 Exercise 19 77

 Exercise 20 77

 Exercise 21 78

 Exercise 22 78

 Exercise 23 78

 Exercise 24 79

 Exercise 25 79

 Exercise 26 80

 Exercise 27 80

Exercise 28	80
Exercise 29	80
Exercise 30	80
Exercise 31	81
Exercise 32	82
Exercise 33	82
Exercise 34	83
Exercise 35	83
Exercise 36	83
Exercise 37	84
Exercise 38	84
Exercise 39	84
Exercise 40	85
Exercise 41	86
Exercise 42	87
Exercise 43	89
Exercise 44	89
Chapter 2: Programming Essentials	91
Writing M/MLX-Files	91
How to Create an M/MLX-File.....	94
Warnings in Scripts	94
Errors in Scripts.....	98
Cell Mode.....	113
Debugging Mode	117
M-Lint Code Check	118
Code Profiling	118
Dependency Report	119
P-Codes	119
Some Remarks About Scripts/M/MLX-Files	121

TABLE OF CONTENTS

Display and Printout Operators: display, sprintf, and fprintf.....	121
Example 1	122
Example 2.....	123
fprintf().....	124
The if, else, elseif, end and switch, case, end Control Statements.....	130
Example 1	132
Example 2.....	133
Example 3.....	135
Example 4.....	140
Loop Control Statements: while, for, continue, break, and end	141
Example 1	142
Example 2.....	143
Example 3.....	147
Example 4.....	150
Example 5.....	151
Example 6.....	154
Example 7	157
Example 8.....	158
Memory Allocation.....	159
Example 9.....	160
Example 10.....	162
Example 11	163
Example 12.....	165
Example 13.....	167
Example 14.....	172
Example 15.....	175
Example 16.....	176
Symbol References in Programming	178
Asterisk	179
At (@) Sign.....	179
Colon	180

Comma	180
Curly Brackets	180
Dollar Sign	181
Dot	181
Dot-Dot	181
Dot-Dot-Dot (Ellipsis)	181
Parentheses	182
Percent	182
Semicolon	183
Single Quotes	184
Slash and Backslash	184
Square Brackets	184
Function Files	185
Example 1	187
Example 2	189
Example 3	190
Example 4	191
Most Common Errors with Function Files	194
Varying Number of Inputs and Outputs	196
Nested and Sub-Functions of Function Files	212
Function Files Within M-Files	215
Summary of Scripts and Function Files	217
Inline Functions	218
Example 1	218
Example 2	219
Example 3	219
Anonymous Functions with Handles (@)	220
Example 1	220
Example 2	221
Example 3	221
How to Improve Script Performance	222

Chapter 3: Graphical User Interface Model Development..... 223

GUIDE 223

 Example 1: Building a 2D Plot..... 228

 Example 2: Adding Functionality 235

 Example 3: Solving a Quadratic Equation..... 238

GUI Dialogs and Message Boxes..... 246

 Error Dialog..... 247

 Warning Message..... 247

 F1 Help/Message Box..... 247

 General Syntax 249

 Input Dialog 250

 Question Dialog 250

Chapter 4: MEX Files, C/C++, and Standalone Applications 259

Verifying Compilers 260

Generating C Code 260

Creating MEX Files from Other Languages 267

Building Standalone Applications 268

Chapter 5: Simulink Modeling Essentials 275

Simulink Modeling 275

 Example: Arithmetic Calculations 277

 Example: Modeling Simple Arithmetic Operations..... 281

 Performing Matrix Operations 285

 Computing Values of Functions 290

 Input/Output Signals from/to the MATLAB Workspace 297

 Simulating a Mechanical System 300

 Working with a Second Order Differential Equation 308

Simulink Model Analysis and Diagnostics..... 313

 Code Generation 314

 Model Advisor 315

Summary..... 321

Exercises for Self-Testing	321
Exercise 1	321
Exercise 2	321
Exercise 3	322
Exercise 4	322
Exercise 5	323
Exercise 6	324
Exercise 7	324
Exercise 8	325
Exercise 9	326
Exercise 10	327
Exercise 11	328
Exercise 12	329
Exercise 13	330
Exercise 14	331
Exercise 15	331
Exercise 16	332
Exercise 17	332
Exercise 18	333
Exercise 19	333
Exercise 20	334
Exercise 21	334
Exercise 22	335
Exercise 23	335
Exercise 24	336
Exercise 25	337
Exercise 26	337
Exercise 27	338
Exercise 28	338
Exercise 29	339
Exercise 30	340
Exercise 31	341

TABLE OF CONTENTS

Exercise 32	341
Exercise 33	342
Exercise 34	342
Exercise 35	342
Chapter 6: Plots and Data Visualization	343
Basics of Plot Building	343
PLOT()	344
Example 1: Plotting Two Rows of Data	344
Example 2: Plotting Function Values.....	345
Example 3: Building a Histogram.....	347
TITLE, XLABEL, YLABEL, AXIS, GRID, and LEGEND	348
Example 4: Plotting a Unit Circle with Plot Tools	349
LINE and MARKER Specifiers.....	350
Example 5: Plotting Sine Function Values with Plot Tools.....	353
Example 6: Plotting Sine Function Values with Plot Tools.....	355
Plot Two Data Sets in Two Y–Y Axes.....	356
Example 7: Plotting Two Function Values on Y-Y Axes	356
Sub-plots.....	359
Example 8: Building Sub-Plots of Functions.....	359
LEGEND	361
HOLD	361
Example 9: Plotting a Few Function Values in One Plot.....	361
Example 10: Plotting Function Values with Different Line Markers and Colors	362
EZPLOT, FPLOTT, and FIMPLICIT with Function Handles (@)	364
Example 11: Plotting a Mathematical Expression with ezplot()	364
GTEXT, TEXT, and GINPUT	365
Example 12: Locate and Display Minimum Values of a Function Plot in a Plot Figure	366
Axis Ticks and Tick Labels	368
Example 13: Display X-Axis Tick Labels	369
Figure Handles	370
Example 14: Working with Figure Handles	370

3D Surface Plots	372
Example 15: Creating a 3D Pie Plot with <code>pie()</code>	372
Example 16: Creating a 3D Surface Plot with <code>ezsurf()</code>	373
Example 17: Creating a 3D Mesh Plot with <code>ezmesh()</code>	374
Example 18: Creating a 3D Surface-Contour Plot with <code>ezsurf()</code>	374
Example 19: Creating a 3D Plot of an Electric Potential Field	375
Example 20: Creating 3D Plots with <code>waterfall()</code> and <code>ribbon()</code>	378
3D Line Plots and Animations	380
Example 21: Building 3D Line Plots and Animated 3D Line Plots with <code>plot3()</code> , <code>comet3()</code> , and <code>ezplot3()</code>	380
Animated Plots	382
Example 22: Building an Animated Plot with <code>getframe()</code>	382
Example 23: Building an Animated Plot with <code>drawnow</code>	383
Example 24: Building an Animated Plot with <code>drawnow</code>	383
Example 25: Building an Animated Plot of a Projectile with <code>getframe()</code>	384
Summary	385
Exercises for Self-Testing	386
Exercise 1	386
Exercise 2	387
Exercise 3	387
Exercise 4	387
Exercise 5	388
Exercise 6	389
Exercise 7	389
Exercise 8	389
Exercise 9	390
Exercise 10	390
Exercise 11	390
Exercise 12	391
Exercise 13	391
Exercise 14	391
Exercise 15	391

TABLE OF CONTENTS

Exercise 16 392

Exercise 17 392

Exercise 18 393

Exercise 19 393

Exercise 20 394

Exercise 21 394

Exercise 22 394

Exercise 23 395

Exercise 24 395

Exercise 25 396

Chapter 7: Linear Algebra 397

Introduction to Linear Algebra..... 397

Matrix Properties and Operators 399

 Simulink Blocks for Matrix Determinant, Diagonal Extraction, and Transpose 400

 Matrix Inverse or Inverse Matrix..... 404

 Simulink Blocks for Inverse Matrix..... 405

Matrix Operations..... 428

 Example: Performing Matrix Operations..... 431

Standard Matrix Generators..... 440

Vector Spaces 445

 Polynomials Represented by Vectors..... 447

 Simulink Model Based Solution of Polynomials 449

Eigen-Values and Eigen-Vectors 451

Matrix Decomposition 454

 QR Decomposition 454

 LU Decomposition..... 457

 Cholesky Decomposition 460

 Schur Decomposition 464

 Singular Value Decomposition (SVD) 467

Logic Operators, Indexes, and Conversions 469

Logical Indexing	470
Example: Logical Indexing to locate and Substitute Elements of A Matrix.....	473
Conversions.....	475
Summary.....	480
References.....	481
Exercises for Self-Testing	482
Exercise 1	482
Exercise 2	483
Exercise 3	483
Exercise 4	484
Exercise 5	484
Exercise 6	485
Exercise 7	485
Exercise 8	485
Exercise 9	486
Exercise 10	486
Exercise 11	486
Exercise 12	487
Exercise 13	487
Exercise 14	487
Exercise 15	487
Exercise 16	488
Exercise 17	488
Exercise 18	488
Exercise 19	488
Exercise 20	489
Exercise 21	489
Exercise 22	489
Exercise 23	489
Exercise 24	489
Exercise 25	490

TABLE OF CONTENTS

Chapter 8: Ordinary Differential Equations 491

Classifying ODEs 492

 Example 1: Unconstrained Growth of Biological Organisms..... 493

 Example 2: Radioactive Decay..... 493

 Example 3: Newton’s Second Law..... 494

Analytical Methods..... 495

 DSOLVE 496

 Example 1: Using dsolve..... 496

 Example 2: Plotting from dsolve 496

 Example 3: Adding an Unspecified Parameter..... 498

Second-Order ODEs and a System of ODEs 499

 Example 1: dsolve with a Second-Order ODE..... 499

 Example 2: System ODEs..... 500

 Example 3: Unsolvable Solutions Using dsolve 501

 Example 4: Computing an Analytical Solution 502

 Example 5: An Interesting ODE 502

 Example 6: Responding to No Analytical Solution 503

Laplace Transforms..... 504

 Example 1: First Laplace Transform 505

 LAPLACE/ILAPLACE..... 506

References..... 516

Index..... 517

About the Author



Sulaymon Eshkabilov, PhD, is currently a visiting professor at the Department of Agriculture and Biosystems, North Dakota State University, USA. He obtained an ME diploma from the Tashkent Automobile Road Institute, an MSc from Rochester Institute of Technology, NY, USA, and a PhD from Cybernetics Institute of Academy Sciences of Uzbekistan in 1994, 2001, and 2005, respectively. He was an associate professor at the Tashkent Automobile Road Institute from December 2006 until January 2017. He held visiting professor and researcher positions at Ohio University, USA, during the 2010/2011 academic year and at Johannes Kepler University,

Austria from January-September, 2017. He teaches the following courses: “MATLAB/Simulink Applications for Mechanical Engineering and Numerical Analysis” and “Modeling of Engineering Systems” to undergraduate students, “Advanced MATLAB/Simulink Modeling” seminar/class, “Control Applications,” “System Identification,” and “Experimentation and Testing with Analog and Digital Devices” to graduate students.

His research interests are mechanical vibrations, control, mechatronics, and system dynamics. He developed simulation and data analysis models for vibrating systems, autonomous vehicle control, and studies of mechanical properties of bones. He has authored two books devoted to MATLAB/Simulink applications for mechanical engineering students and numerical analysis. He has worked as an external academic expert in the European Commission to assess academic projects in 2019.

About the Technical Reviewers



Dr. Joseph Mueller specializes in control systems and trajectory optimization. For his doctoral thesis, he developed optimal ascent trajectories for stratospheric airships. His active research interests include robust optimal control, adaptive control, applied optimization and planning for decision support systems, and intelligent systems to enable autonomous operations of robotic vehicles. Prior to joining SIFT in early 2014, Dr. Mueller worked at Princeton Satellite Systems for 13 years. In that time, he served as the principal investigator for eight small business innovative research contracts for NASA, the Air Force, the Navy, and MDA. He has developed algorithms for optimal guidance and control of both formation flying spacecraft and high altitude airships and developed a course of action planning tool for DoD communication satellites. In support of a research study for the NASA Goddard Space Flight Center in 2005, Dr. Mueller developed the Formation Flying Toolbox for MATLAB, a commercial product that is now used at NASA, ESA, and several universities and aerospace companies around the world. In 2006, Dr. Mueller developed the safe orbit guidance mode algorithms and software for the Swedish Prisma mission, which has successfully flown a two-spacecraft formation flying mission since its launch in 2010. Dr. Mueller also serves as an adjunct professor in the Aerospace Engineering & Mechanics Department at the University of Minnesota, Twin Cities campus.

ABOUT THE TECHNICAL REVIEWERS



Karpur Shukla is a research fellow at the Centre for Mathematical Modeling at FLAME University in Pune, India. His current research interests focus on topological quantum computation, nonequilibrium, and finite-temperature aspects of topological quantum field theories, and applications of quantum materials effects for reversible computing. He received an M.Sc. in physics from Carnegie Mellon University, with a background in theoretical analysis of materials for spintronics applications as well as Monte Carlo simulations for the renormalization group of finite-temperature spin lattice systems.

Acknowledgments

I would like to express my special gratitude to the technical reviewers of this book—Dr. Joseph Mueller and Mr. Karpur Shukla—for their very thorough work while reviewing the context and code of the book. Without their critical insights and corrections many points along the way, the book would not be as good as it is now. I would also like to express my special gratitude to Mark Powers for his timely and well planned correspondence throughout this book project.

My cordial gratitude goes to my mother, for her limitless support and love. Up until the last point of this book, she has kept checking my progress.

I would like to thank my wife, Nigora. Without her great support, I would not be able to take up the challenging task of writing this book. I have spent so much time over the weekends in my office writing and editing the book. In addition, I would like to thank our children—Anbara, Durdona, and Dovud—for being such delightful people and for being the inspiration for writing this book.

Introduction

This book is aimed at beginner-level learners of MATLAB/Simulink packages. It covers the essential, hands-on tools and functions of the MATLAB and Simulink packages, and explains them via interactive examples and case studies. The main principle of the book is “learning by doing” and it progresses from simple to complex. It contains dozens of solutions and simulation models via M/MLX files/scripts and Simulink models, which help you learn the programming and modeling essentials. Moreover, there are many recommendations for avoiding pitfalls related to programming and modeling aspects of MATLAB/Simulink packages.

“Beginning MATLAB and Simulink” explains various practical issues of programming and modeling in parallel by comparing programming tools of MATLAB and blocks of Simulink. After studying this book, you'll be proficient at using MATLAB/Simulink packages. You can apply the source code and models from the book's examples as templates to your own projects in data science, numerical analysis, modeling and simulation, or engineering.

Essential learning outcomes of the book include:

- Getting started using MATLAB and Simulink
- Performing data analysis and visualization with MATLAB
- Programming essentials of MATLAB and core modeling aspects of Simulink, and how to associate scripts and models of MATLAB and Simulink packages
- Developing GUI models and standalone applications in MATLAB
- Working with integration and numerical root-finding methods in MATLAB and Simulink
- Solving differential equations in MATLAB and Simulink
- Applying MATLAB for data analysis and data science projects

The book contains eight logically interlinked chapters.

- Chapter 1 is dedicated to introducing the MATLAB environment and MATLAB recognized data types, including numeric, cell, structure, character, logical, function handle, and table.
- Chapter 2 covers most of the essential programming tools and functions, such as `for ... end` and `while ... end` loop operators, `if...elseif...else...end` condition operators, symbol referencing, and most common errors and warnings. It also covers M-file debugging tools and options in MATLAB. Moreover, this chapter addresses MATLAB-specific programming tools, including function files, function handles, and display operators, such as `disp()`, `display`, `fprintf`, and `sprint`.
- Chapter 3 covers GUI development and how to write and edit GUI model callback functions.
- Chapter 4 addresses the issues of how to develop MEX files, C/C++ code, and standalone applications from the existing M-files.
- Chapter 5 is dedicated to Simulink modeling essentials. The examples in this chapter cover matrix operations, computing function values, modeling, and solving ordinary differential equations. Moreover, it covers issues around how to associate Simulink models with MATLAB scripts.
- Chapter 6 is devoted to data visualization issues, such as building 2D and 3D plots and animated plots in MATLAB.
- Chapter 7 is dedicated to matrix algebra and array operations. It addresses solving systems linear equations, eigen-value problems, matrix decompositions, matrix and vector operations, and conversions of arrays and strings via examples solved in MATLAB and Simulink in parallel.
- Chapter 8 covers some essential aspects of solving ordinary differential equations analytically and using MATLAB's built in functions and commands.

All of the source code (scripts, M/MLX/MAT files, Simulink models, SLX/MDL files, C code, MEX-files, and installation *.exe files) discussed in the book are available to readers via a Download Source Code button on the book's [apress.com](http://www.apress.com) product page and accessible via the Download Source Code link located at www.apress.com/9781484250600.

Note The scripts in the book may not always be the best solutions to the given problems, but this is done intentionally in order to emphasize methods used to improve them. In some other cases, it is found to be the most appropriate solution to the author's best knowledge. Should I spot better alternative solutions to exercises, I will publish them via MathWorks' MATLAB Central User Community's file exchange, via my file exchange link there (under my name).

No matter how hard we worked to proofread this book, it is inevitable that there will be some typographical errors that might slip through and will appear in print.

September 2019

Sulaymon Eshkabilov